StrokeAegis AI

AI-Powered Stroke Reporting & Integrated Solutions
Clinical Decision Support Platform (Multi-Agent Architecture)

REPORT ID: SA-2026-FATIMA-BLEED
DATE: 2026-09-24 10:48
STATUS: VALIDATED AI ASSESSMENT
ACCESS: Confidential Medical Documentation

1. Patient Demographic & Vitals Profile

Patient Name: Fatima Al-Zahrani	**MRN:** MRN-202609-002	**Attending Neurologist:** Dr. Sarah Al-Mansoor, MD	**Record Status:** Active Patient File
Gender: Female	**Age:** 59 Years	**BMI:** 32.1 kg/m²	**Smoking:** never smoked
Hypertension: Positive (Yes)	**Cardiac Disease:** Negative	**Average Glucose:** 195.4 mg/dL	**Residence:** Urban

2. Clinical Machine Learning Stroke Risk Stratification (StrokePrediction-V2)

Predicted Probability: 65.2% | **Risk Tier:** HIGH DANGER TIER | **Decision Threshold: 49.0%**

Clinical Risk Multipliers: Acute Intracerebral Hemorrhage, Stage 2 Hypertension

Model Rationale: High-risk presentation due to acute hemorrhagic findings and severe systemic hypertension.

3. Diagnostic CT Neuroimaging & Explainability (EfficientNetV2-S + GeM)

Gatekeeper Safety Status: Passed Gatekeeper (99.2% Brain CT confidence)
Primary CT Classification: BLEEDING (91.5% Confidence)
Status Banner: N/A
Lesion Localization: Right Basal Ganglia Hemorrhage
Hotspot Lesion Coverage: 0.0% of Brain ROI

No lesion overlay required

4. AI Clinical Decision Support (CrewAI Guideline-Grounded Multi-Agent)

4.1 Hyperacute Emergency Stroke Management Specialist

Acute Summary: N/A
Reperfusion Considerations: N/A
Acute Blood Pressure Target: N/A

4.2 Guideline-Based Stroke Prevention Specialist

Vascular Risk Interpretation: N/A
Blood Pressure Target: N/A
Glycemic & Metabolic Target: N/A
Recommended Lifestyle Modifications:

5. Medical Safety & Clinical Validation Audit

Audit Outcome: PASSED | **Safety Verification:** PASSED | **Non-Autonomous Stance Verified:** True
Clinical Contradictions Detected: 0 | **Critical Safety Alerts:** 0

6. Integrated Clinical Action Plan & Executive Summary

Acute hyperdense intraparenchymal hemorrhage detected in right basal ganglia. Urgent neurocritical intervention required.

- **Stat neurosurgical consult for hematoma evaluation.**

- **Immediate systolic BP reduction < 140 mmHg.**
- **Reversal of any antiplatelet or anticoagulant agents.**

7. Traceable Medical Guideline References (Medical RAG Knowledge Base)

CLINICAL DISCLAIMER: StrokeAegis AI is an AI-powered clinical decision-support system intended to assist qualified healthcare professionals. It does not replace clinical judgment, diagnosis, or physician decision-making and does not independently prescribe or order treatment.

Attending Physician Evaluation:

Dr. Sarah Al-Mansoor, MD
Status: **Pending Physician Sign-Off**
Date: 2026-09-24